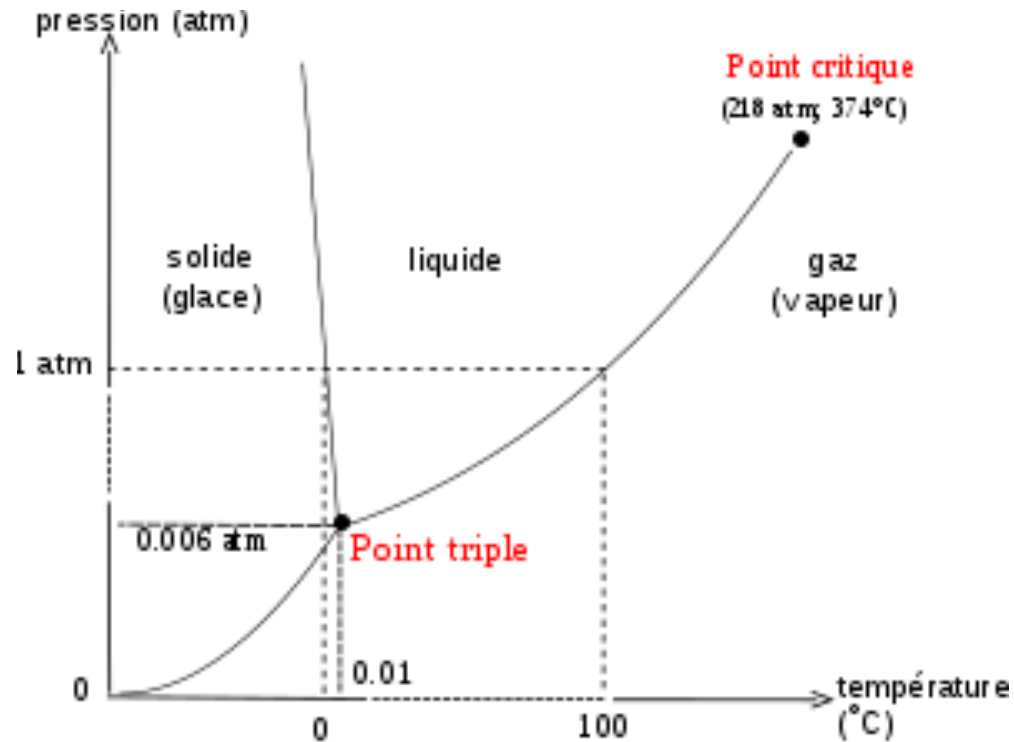
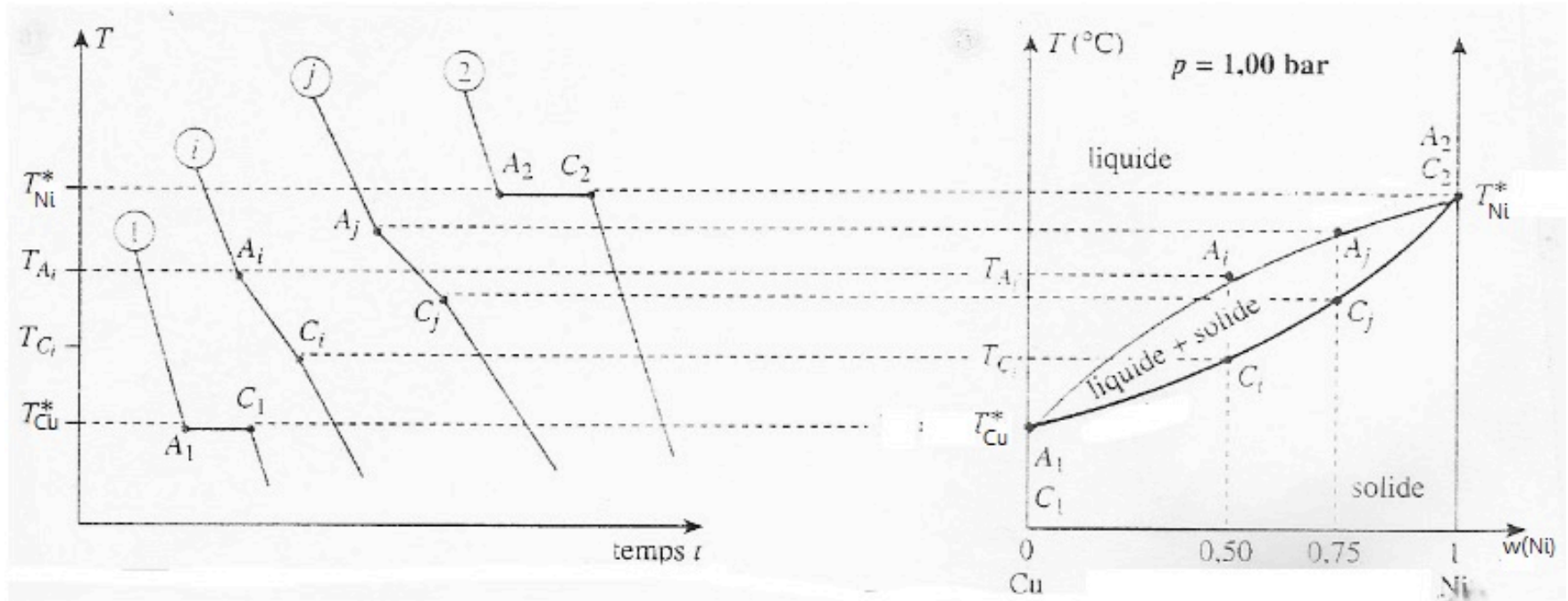


Du corps pur au mélange binaire

Diagramme de phase de l'eau



Construction d'un diagramme binaire phases miscibles



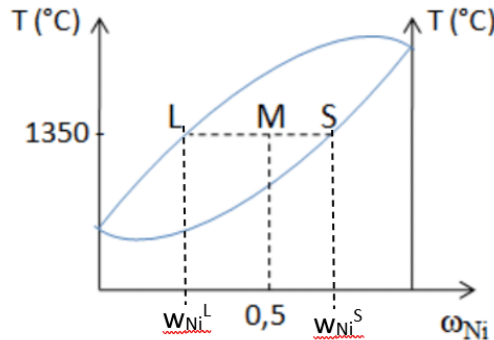
Théorème des moments chimiques

- $m_{\text{Ni}} = m_{\text{Ni}}^{\text{liq}} + m_{\text{Ni}}^{\text{S}} = w_{\text{Ni}}^{\text{liq}} \cdot m^{\text{liq}} + w_{\text{Ni}}^{\text{S}} \cdot m^{\text{S}}$
- $m_{\text{Ni}} = w_{\text{Ni}} \cdot m_{\text{tot}} = w_{\text{Ni}} \cdot (m^{\text{liq}} + m^{\text{S}})$

$$\left. \begin{array}{l} m_{\text{Ni}} = m_{\text{Ni}}^{\text{liq}} + m_{\text{Ni}}^{\text{S}} = w_{\text{Ni}}^{\text{liq}} \cdot m^{\text{liq}} + w_{\text{Ni}}^{\text{S}} \cdot m^{\text{S}} \\ m_{\text{Ni}} = w_{\text{Ni}} \cdot m_{\text{tot}} = w_{\text{Ni}} \cdot (m^{\text{liq}} + m^{\text{S}}) \end{array} \right\} w_{\text{Ni}} \cdot (m^{\text{liq}} + m^{\text{S}}) = w_{\text{Ni}}^{\text{liq}} \cdot m^{\text{liq}} + w_{\text{Ni}}^{\text{S}} \cdot m^{\text{S}}$$

$$m^{\text{S}} \cdot (w_{\text{Ni}} - w_{\text{Ni}}^{\text{S}}) + m^{\text{liq}} \cdot (w_{\text{Ni}} - w_{\text{Ni}}^{\text{liq}}) = 0$$

Cas du mélange
cuivre-Nikel



$$w_{\text{Ni}}^{\text{S}} = 0,8$$
$$w_{\text{Ni}}^{\text{L}} = 0,4$$

Avec $m_{\text{tot}} = 100\text{g}$

Diagramme binaire phases solides non miscibles (Pb-Sn)

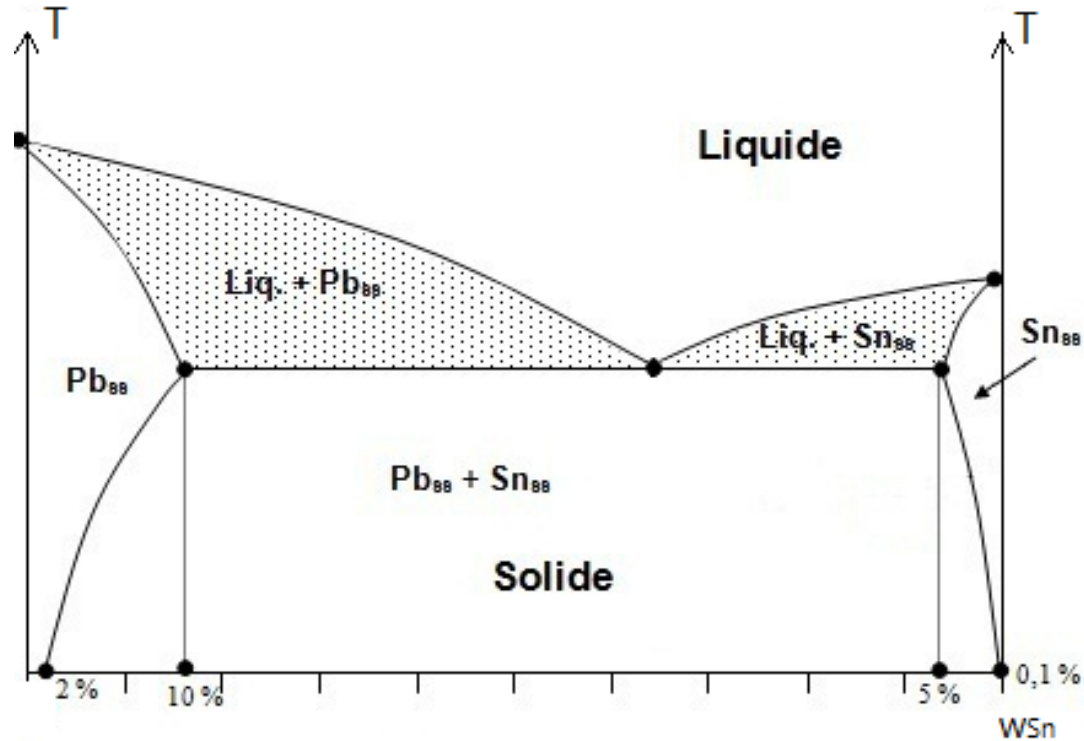


Diagramme binaire et composé défini (Mg-Si)

